SECTIONS OF THE STRUCTURE SHEAF OF A THICKENING SURJECT ONTO THOSE OF THE CLOSED SUBSPACE ON AFFINE ETALE OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-more-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the first line of the pinned chapter file is \input{preamble}, and the native preamble was not staged with the source, so no preamble text lives inside the chapter. The retained preamble used here is the project's single Stacks-wide preamble (the same 217-line / 8116-byte file that the already published chow and etale-cohomology cards use), copied mechanically line for line from sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author supplies two proof environments for this lemma, the second titled 'Proof for finite order thickenings' and introduced in the text as the case most used in practice (proof lines 1696-1706 and 1708-1727). Both are counted, in the order the author gives them, which is why this card carries two proof spans.

Allowed standard prerequisites (editorial).

- Etale localization of thickenings of algebraic spaces. Same chapter, Theorem labelled theorem-topological-invariance; invoked at line 1698. Produces the etale U' -> X' with U = X x_{X'} U' for the affine object U of X_{spaces,etale}.
- Affineness descends along the thickening. Limits of Spaces, Lemma labelled spaces-limits-lemma-affine; invoked at lines 1699-1700. Shows that U' is an affine scheme, so its global sections can be compared with those of U.
- The kernel of a first order thickening is a quasi-coherent module on the closed subspace. Morphisms of Spaces, Lemma labelled spaces-morphisms-lemma-i-star-equivalence; invoked at lines 1713-1716. Identifies the square-zero ideal I with a quasi-coherent O_X-module in the second proof.
- Etale cohomology of a quasi-coherent module on an affine scheme vanishes in positive degrees. Descent, Proposition labelled descent-proposition-same-cohomology-quasi-coherent and Cohomology of Schemes, Lemma labelled coherent-lemma-quasi-coherent-affine-cohomology-zero; invoked at lines 1722-1726. Kills H^1_etale(U, I), which is what makes the section map surjective.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Navigation pointer for \ref{coherent-lemma-quasi-coherent-affine-cohomology-zero} used inside the counted span (line 1726); the argument that consumes it is retained in the span.

Source reference. Original lemma 03DW: sources/stacks-chow/tags.txt, source lines 4435–4435. Navigation pointer for \ref{descent-proposition-same-cohomology-quasi-coherent} used inside the counted span (line 1724); the argument that consumes it is retained in the span.

Source reference. Original lemma 05ZM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1554–1554. Navigation pointer for \ref{equation-fundamental-equivalence} used inside the counted span (line 1693); the argument that consumes it is retained in the span.

Source reference. Original lemma 07VS: sources/stacks-chow/tags.txt, source lines 10017–10017. Navigation pointer for \ref{spaces-limits-lemma-affine} used inside the counted span (line 1699); the argument that consumes it is retained in the span.

Source reference. Original lemma 04CJ: sources/stacks-chow/tags.txt, source lines 5693–5693. Navigation pointer for \ref{spaces-morphisms-lemma-i-star-equivalence} used inside the counted span (line 1715); the argument that consumes it is retained in the span.

Source reference. Original lemma 05ZH: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1316–1316. Navigation pointer for \ref{theorem-topological-invariance} used inside the counted span (line 1698); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $X \subset X'$ be a thickening of algebraic spaces over S. Let U be an affine object of $X_{spaces,étale}$. Then*

$$\Gamma(U, \mathcal{O}_{X'}) \to \Gamma(U, \mathcal{O}_X)$$

is surjective where we think of $\mathcal{O}_{X'}$ as a sheaf on $X_{spaces,étale}$ via (05ZM).

Proof. Let $U' \to X'$ be the étale morphism of algebraic spaces such that $U = X \times_{X'} U'$, see Theorem 05ZH. By Limits of Spaces, Lemma 07VS we see that U' is an affine scheme. Hence $\Gamma(U, \mathcal{O}_{X'}) = \Gamma(U', \mathcal{O}_{U'}) \to \Gamma(U, \mathcal{O}_U)$ is surjective as $U \to U'$ is a closed immersion of affine schemes. Below we give a direct proof for finite order thickenings which is the case most used in practice. □

Proof for finite order thickenings. We may assume that $X \subset X'$ is a first order thickening by the principle explained above. Denote $\mathcal{I}$ the kernel of the surjection $\mathcal{O}_{X'} \to \mathcal{O}_X$. As $\mathcal{I}$ is a quasi-coherent $\mathcal{O}_{X'}$-module and since $\mathcal{I}^2 = 0$ by the definition of a first order thickening we may apply Morphisms of Spaces, Lemma 04CJ to see that $\mathcal{I}$ is a quasi-coherent $\mathcal{O}_X$-module. Hence the lemma follows from the long exact cohomology sequence associated to the short exact sequence

$$0 \to \mathcal{I} \to \mathcal{O}_{X'} \to \mathcal{O}_X \to 0$$

and the fact that $H^1_{étale}(U, \mathcal{I}) = 0$ as $\mathcal{I}$ is quasi-coherent, see Descent, Proposition 03DW and Cohomology of Schemes, Lemma 01XB. □